## Chapter 3: Framework and Methodology

Proposal outline for team and adviser review. The conceptual framework will explain the relationships to be examined; the methodological framework will explain the sequence of research procedures. The department headings will be retained, with the methodological framework placed under Section 3.2.

Tag key:

- LOCKED will identify requirements already stated in the research brief.
- [DECISION PENDING: ...] will identify choices for the team.
- [NEEDS VERIFICATION] will identify unconfirmed facts or requirements.

Official documentation was checked on 11 October 2026. Verified facts will not imply that a dataset, version, threshold, or software implementation has been selected.

Citation numbering will preserve the existing source [47]. New sources will provisionally begin at [57], after the highest citation appearing in Chapter I. The supplied bibliography ends at [51], so the missing [56] and any reserved entries [52]–[55] will require reconciliation before submission.

### 3.1 Conceptual Framework

**Purpose:** The section will explain how historical node attributes and alternative relational representations will be examined in relation to 2025 LST estimation, and will connect the comparisons to RQ1–RQ3.

- LOCKED: The framework will distinguish historical environmental attributes, graph adjacency, estimated LST, and subsequently derived SUHI intensity. Road adjacency will be treated as a candidate relational representation whose usefulness will be tested, rather than a physical heat-transfer pathway.

- LOCKED: Historical LST, NDVI, and NDBI for 2019–2024 will provide temporal node attributes. An LSTM encoder will produce temporal representations; GraphSAGE layers will aggregate information according to the specified adjacency; the regression output will estimate 2025 LST. GraphSAGE will provide a methodological basis for neighborhood aggregation [47].

- LOCKED: RQ1 will concern the accuracy of road-connected LSTM-GraphSAGE at held-out urban road nodes. RQ2 will concern performance changes relative to self-loops only and a spatial-proximity graph. RQ3 will concern comparison with Random Forest using the same historical predictors and held-out observations.

- LOCKED: Configurations 1–3 will share input features, layer dimensions, and training procedure. Their comparison will examine the contribution of the chosen adjacency representation under the documented experiment; it will not establish causality or presume improvement.

- LOCKED: SUHI intensity will be a downstream interpretive quantity: estimated urban-node LST minus the common target-year non-urban reference LST. SUHI intensity will not be a second model output or a separately evaluated target.
- [DECISION PENDING: D1 conceptual framework presentation]: The team will confirm whether Section 3.1 will use an input-process-output diagram, a relationships diagram, or a theory-based presentation. The research brief leaves the department's expected style unfilled; no style will be assumed.
- [DECISION PENDING: D2 literature linkage]: The team will confirm the final Chapter II section titles and theoretical references supporting temporal encoding, graph aggregation, spatial evaluation, and reference-based SUHI interpretation. The supplied Chapter II currently contains Land Surface Temperature and Surface Urban Heat Islands and Landsat-Based Environmental Observations, both numbered 2.1.1.

### 3.2 Methodology

**Purpose:** The section will specify the comparative research design, data workflow, model construction, leakage controls, evaluation, mapping, and reproducibility procedures.

#### 3.2.1 Methodological Framework

**Purpose:** This subsection will present the procedural framework separately from the conceptual relationships in Section 3.1.

- LOCKED: The workflow will show data acquisition, quality screening and annual compositing, mask definition, road graph construction, raster-to-node assignment, spatial partitioning, model training, common LST evaluation, and subsequent SUHI derivation and mapping [57]–[65].
- LOCKED: Historical predictors and the 2025 LST target will follow separate processing branches. The target-year reference LST will enter only the downstream SUHI mapping branch; it will not enter model predictors or adjacency construction.
- LOCKED: The workflow will include decision gates for masks, spatial support, graph construction, split design, and model settings before final test evaluation. It will identify which transformations will be fitted using training data and subsequently applied to validation and test observations.
- [DECISION PENDING: D3 workflow figure detail]: The team will confirm whether the methodological framework will use a single overview figure or an overview with a detailed leakage-control panel. The figure will display dependencies without selecting unresolved parameters.

#### 3.2.2 Research Design and Analytical Units

**Purpose:** This subsection will define the retrospective comparative estimation experiment and the interpretation of its observations.

- LOCKED: The study will use quantitative supervised regression to estimate 2025 LST from the six annual historical observations per node. It will compare four configurations on the same held-out observations and evaluate LST using RMSE, MAE, and $R^2$.
- LOCKED: The reporting locations will be road-associated locations represented by graph nodes. Node densification will not be treated as additional independent satellite measurements. Conclusions will concern the evaluated annual-composite task, rather than operational forecasting or seasonal and diurnal behavior.

- [DECISION PENDING: D4 training population and graph extent]: The team will specify whether training and message passing will use urban nodes only or an expanded graph with non-urban context nodes. Urban held-out evaluation will remain common across configurations; target availability and the eligibility of context nodes will be documented.

### 3.2.3 Study Area and Urban and Reference Masks

**Purpose:** This subsection will define eligible urban nodes and terrestrial non-urban reference pixels independently of prediction errors.

- LOCKED: Urban nodes will be identified through an objective land-cover dataset rather than a manually drawn urban boundary. A non-urban reference mask will be defined independently of model errors, with boundary and eligibility criteria fixed before final evaluation. Water surfaces will be excluded from the reference.
- [DECISION PENDING: D5 urban mask]: The team will select the land-cover dataset, product year, version, built-up class or fraction threshold, spatial continuity rule, and boundary-pixel treatment. If ESA WorldCover will be used, its documentation will support product interpretation; a historical map will not automatically represent 2025 land cover [60], [61].
- [DECISION PENDING: D6 reference geography and cover]: The team will define the reference boundary, distance or buffer rules, admissible terrestrial land-cover classes, mixed-pixel thresholds, and urban-edge exclusions. Reference pixels will remain eligible even where they will fall outside the modeled urban graph.
- [DECISION PENDING: D7 elevation comparability]: The team will select an elevation dataset and determine whether elevation matching, a restriction to comparable elevation ranges, or documented stratification will be used. The purpose will be to limit confounding of the urban–reference contrast by mountainous reference surfaces.
- [DECISION PENDING: D8 spatial coordinate system]: The team will select the administrative boundary snapshot, projected coordinate system for distance calculations, raster grid, and alignment rules. These choices will be recorded before distance-based graph and split construction.

### 3.2.4 Data Acquisition and Provenance

**Purpose:** This subsection will specify data sources, temporal coverage, retrieval procedures, and traceable input records.

- LOCKED: Landsat 8 Collection 2 Level-2 annual median composites will provide historical LST, NDVI, and NDBI for 2019–2024 and observed LST for 2025. Landsat 9 will be excluded. Product structure and QA information will be documented using USGS guidance [57], [59].
- LOCKED: OpenStreetMap road data will be obtained through OSMnx. Nodes will represent intersections and intermediate road points; adjacency will represent retained road connections [62].
- [DECISION PENDING: D9 acquisition platform and provenance]: The team will select download or cloud-processing services, scene eligibility, retrieval dates, and storage formats. Scene identifiers, acquisition dates, processing metadata, and the final imagery inventory will be retained.
- [NEEDS VERIFICATION]: The exact delivered product identifiers, processing revisions, availability of temperature bands, and software releases will be checked against the actual downloaded assets and installed environment. Documentation versions observed online will not be reported as installed versions.

### 3.2.5 Satellite Preprocessing and Annual Composites

**Purpose:** This subsection will define valid observations, physical-unit conversion, spectral indices, aggregation, and missing-data treatment.

- [DECISION PENDING: D10 quality screening]: The team will specify QA_PIXEL and QA_RADSAT handling, cloud and cloud-shadow rules, cirrus and dilated-cloud treatment, invalid/fill exclusion, temperature uncertainty screening, cloud-distance rules if used, and minimum annual valid-observation requirements [57], [59].
- [DECISION PENDING: D11 scaling implementation and units]: The team will confirm the implementation of product scaling and the reporting temperature unit. Verified USGS guidance will constrain the conversion: ST in kelvin = DN × 0.00341802 + 149.0; surface reflectance = DN × 0.0000275 − 0.2. The implementation will apply valid-data masks and avoid scaling already-scaled values. Celsius conversion, if chosen, will subtract 273.15 after kelvin conversion [58].
- [DECISION PENDING: D12 index and composite order]: The team will specify the exact NDVI and NDBI band equations, denominator handling, and whether annual indices will be medians of scene-level indices or indices computed from annual reflectance composites. This choice will remain consistent across years and nodes [57], [66].
- LOCKED: Annual medians will be formed from valid observations. The 2019–2024 predictor sequence will be kept separate from 2025 target values. The reference value will be the median of valid annual-composite LST pixels in the fixed reference mask for the same year, rather than the median of node samples.
- [DECISION PENDING: D13 missing values]: The team will choose complete-sequence exclusion, training-fitted imputation, or an explicitly documented missingness-aware approach. Missing 2025 targets will not be invented for evaluation. Coverage and retained sample counts will be reported by year and partition.
- [NEEDS VERIFICATION]: Product-specific gaps and retrieval limitations will be documented from the selected assets and USGS guidance. The team will not assume that cloud screening will produce identical effective sampling or uniformly valid coverage across years [57], [66].

### 3.2.6 Road Graph Construction

**Purpose:** This subsection will define a reproducible road topology and the conversion from geographic road geometry to model adjacency.

- [DECISION PENDING: D14 OSM snapshot and road filters]: The team will select the extraction or historical snapshot date, road-type filters, access restrictions, and clipping policy. A later road snapshot will not be assumed to reproduce road conditions throughout 2019–2025 [62].
- [DECISION PENDING: D15 simplification and node spacing]: The team will specify whether intermediate OSM geometry nodes will be retained or intermediate points will be inserted after simplification. Spacing, intersection consolidation, duplicate-node handling, and retained geometry will be documented. OSMnx simplification can remove intermediate geometry nodes, so defaults will not be assumed to satisfy the node definition [62].
- [DECISION PENDING: D16 adjacency conventions]: The team will define directed or undirected adjacency, treatment of one-way roads, parallel edges, self-loops, intersection connectivity, and any edge attributes. Geometric crossings will not automatically be treated as connected intersections.

- [DECISION PENDING: D17 disconnected components]: The team will choose retention of all eligible components, a documented component-size rule, or a restricted main-component analysis. Urban eligibility will not be changed solely to increase connectivity. Coverage changes and isolated nodes will be reported [62].

### 3.2.7 Raster to Node Assignment and Spatial Support

**Purpose:** This subsection will specify how satellite predictors and targets will be associated with road-associated locations and how shared observation support will be tracked.

- [DECISION PENDING: D18 raster assignment]: The team will select containing-pixel or nearest-cell extraction, interpolation, or a documented neighborhood summary. Raster resampling, grid alignment, and mixed-pixel treatment will be explicit. The same rule will be applied across configurations.
- LOCKED: Node spacing will not determine thermal measurement resolution. Landsat 8 TIRS observations will be interpreted using their 100 m native thermal sampling and the delivered product grid; a finer resampled grid will not be treated as independent fine-scale temperature observations [57], [66].
- [DECISION PENDING: D19 shared support grouping]: The team will define source-pixel identifiers or contributor footprints for predictors and targets. Nodes sharing target pixels will not be assigned to different partitions. Interpolated or neighborhood-derived samples will require checks for overlapping contributors; spatial buffers will address dependence beyond identical delivered cells.
- [DECISION PENDING: D20 evaluation weighting]: The team will choose equal-node weighting, equal-source-pixel weighting, or reporting both. The decision will distinguish the estimand at road nodes from the number of independent thermal observations and will control the influence of densely sampled road segments.

### 3.2.8 Model Configurations

**Purpose:** This subsection will define the matched adjacency comparisons and the conventional baseline without assuming a preferred result.

- LOCKED: Configuration 1 will use road-connected LSTM-GraphSAGE; Configuration 2 will use matched LSTM-GraphSAGE with adjacency restricted to self-loops only; Configuration 3 will use matched LSTM-GraphSAGE with a spatial-proximity graph; Configuration 4 will use Random Forest with the same historical predictors [47], [63], [64].
- LOCKED: Configurations 1–3 will share input features, layer dimensions, and training procedure. The Random Forest comparison will concern the complete model relative to a tabular benchmark; it will not independently isolate the effect of road adjacency.
- [DECISION PENDING: D21 proximity graph]: The team will select KNN or another proximity rule. If KNN will be used, k, metric, projected distances, directed or symmetrized edges, ties, maximum-distance restrictions, and graph scope at split boundaries will be specified without reference to final test performance.
- [DECISION PENDING: D22 neural architecture]: The team will specify LSTM dimensions and depth, sequence representation, GraphSAGE depth and dimensions, aggregation, activations, dropout, normalization, and regression head. The six-year sequence will contain the same three historical predictors for each configuration.
- [DECISION PENDING: D23 self-information implementation]: The team will specify explicit self-loop inclusion and root-feature handling consistently across the three neural configurations.

PyTorch Geometric SAGEConv will have a separate root-feature contribution, so an identity adjacency and an empty edge list will not be silently treated as interchangeable [63].

- [DECISION PENDING: D24 Random Forest encoding and settings]: The team will define a lossless tabular encoding of the same historical features, such as year-specific columns, and select tree count, depth, feature sampling, leaf constraints, and tuning budget. Unmatched additional spatial or environmental predictors will not be introduced [64].

### 3.2.9 Spatial Partitioning and Leakage Controls

**Purpose:** This subsection will define what generalization the evaluation will test and will prevent contamination from shared observation support, fitted preprocessing, or target access.

- LOCKED: Train, validation, and test observations will be partitioned spatially. The same held-out observations will be used for all four configurations. Controls will account for shared satellite pixels and spatial dependence.
- [DECISION PENDING: D25 spatial split]: The team will choose spatial blocks, buffered held-out areas, or geographically separated groups. Block or buffer sizes, partition proportions, assignment rules, held-out areas versus held-out nodes, and repetition will be specified before final evaluation. A node-random split will not substitute for the required spatial design.
- [DECISION PENDING: D26 transductive or inductive protocol]: The team will specify whether unlabeled held-out node features and topology will be visible during training. A transductive protocol will explicitly permit defined historical-feature access; an inductive protocol will specify removal of held-out nodes or areas during fitting and their treatment at inference. GraphSAGE's inductive capability will not by itself make an experiment inductive [47].
- [DECISION PENDING: D27 boundary message passing]: The team will define whether cross-partition road and proximity edges will be removed, constrained by buffers, or allowed under a declared transductive protocol. Multi-layer neighborhoods, sampling, and normalization will follow that policy. Historical-feature exchange will be distinguished from prohibited target-label leakage.
- LOCKED: Validation and test 2025 LST will not enter predictor tensors, adjacency construction, imputation fits, or training loss. Training-only transformations will be applied to the held-out partitions. Validation observations will support model selection; final test results will not guide subsequent tuning.
- [DECISION PENDING: D28 preprocessing and test governance]: The team will specify scaler and imputer fitting, validation access, early-stopping selection, checkpoint selection, and whether training plus validation refitting will occur after settings are frozen. Pixel grouping will remain intact during any refit.

### 3.2.10 Training and Model Selection

**Purpose:** This subsection will specify a matched and reproducible optimization procedure with a defined validation role.

- [DECISION PENDING: D29 training settings]: The team will choose loss function, optimizer, learning rate, batch or full-graph mode, neighborhood sampling, epoch limit, early-stopping criterion and patience, regularization, initialization, seeds, and compute budget.
- LOCKED: The three neural configurations will use a shared training procedure. Architectural changes or additional predictors will not accompany the adjacency ablations.
- [DECISION PENDING: D30 tuning policy]: The team will choose fixed shared settings or a common validation-based selection procedure for the neural configurations and define a

comparable Random Forest search budget. The policy will document whether matched final settings or matched search procedures will be required.

### 3.2.11 Performance Evaluation and Statistical Comparison

**Purpose:** This subsection will answer RQ1–RQ3 using common LST observations and a prespecified comparison procedure.

- LOCKED: RMSE, MAE, and $R^2$ will be computed on LST. RMSE and MAE will use the selected temperature unit; $R^2$ will be dimensionless. Metric definitions, sample counts, and any constant-target handling will be documented [65].
- LOCKED: RQ1 will report road-connected LSTM-GraphSAGE performance. RQ2 will report road-connected minus self-loops only and road-connected minus spatial-proximity graph metric differences. RQ3 will report road-connected minus Random Forest differences. Lower error metrics and higher $R^2$ will indicate better estimation, but differences of either sign will be reported.
- [DECISION PENDING: D31 statistical comparison]: The team will choose paired spatial-block resampling, paired comparisons across repeated spatial partitions, or descriptive paired differences without significance testing. The analysis unit, repetitions, uncertainty intervals, significance level if used, and treatment of multiple comparisons will be prespecified. Individual nodes sharing satellite support will not be treated as independent replicates.
- [DECISION PENDING: D32 aggregation and diagnostic reporting]: The team will choose pooled versus partition-level reporting, seed aggregation, metric priority, and residual diagnostics. Variation across random seeds will be distinguished from uncertainty about spatial generalization. No test-based filtering of difficult nodes will be performed.

### 3.2.12 SUHI Derivation and Mapping

**Purpose:** This subsection will specify downstream reference-based interpretation while retaining LST as the sole prediction and evaluation target.

- LOCKED: The study will compute LSTref(2025) as the median of valid annual-composite LST pixels in the fixed terrestrial non-urban reference mask. Estimated SUHI(i,2025) will equal estimated LST(i,2025) − LSTref(2025).
- LOCKED: The reference value will remain common across configurations and any observed contextual SUHI map. Reference pixels will not have to be road nodes. Neither a separate SUHI prediction model nor a separate SUHI accuracy assessment will be introduced.
- [DECISION PENDING: D33 mapping design]: The team will select mapped configurations, symbols, class breaks or continuous scale, and treatment of missing predictions. Maps will identify the year, unit, reference definition, spatial support, and whether predictions will be held-out estimates or subsequent full-coverage outputs. Any interpolation will be identified explicitly.
- LOCKED: Maps will support further site assessment. They will not demonstrate pavement-scale thermal measurements, pedestrian exposure, intervention effectiveness, causal heat transport, or operational heat-health warnings.

### 3.2.13 Software and Reproducibility

**Purpose:** This subsection will define the records needed to reproduce the data preparation, comparisons, and reported results.

- [DECISION PENDING: D34 implementation and reproducibility]: The team will select preprocessing tools, model frameworks, exact installed versions, hardware, environment locking, repository and archival location, data-access manifests, and output formats. OSMnx documentation, PyTorch Geometric documentation, and scikit-learn documentation will inform implementation without fixing an unselected stack [62]–[65].
- LOCKED: Reproducibility records will include source dates, imagery identifiers, mask definitions, graph rules, retained-node and pixel counts, split membership, leakage checks, settings, seeds, checkpoints, and evaluation outputs.
- [NEEDS VERIFICATION]: Selected software licenses, processing-service terms, and redistribution conditions for derived assets will be checked against the exact tools and artifacts used. Data-product permissions will not automatically establish the permissions for an entire processing platform.

### 3.3 Ethical Consideration

**Purpose:** The section will document lawful public-data use, responsible interpretation, transparent reporting, and institutional requirements.

- LOCKED: The study will use public geospatial and satellite data and will involve no human participants or personal data. It will not collect identities, individual movement histories, or personal heat-health information.
- LOCKED: Landsat data will be acknowledged as USGS data in the public domain. Public-domain status was confirmed through official USGS guidance [68]. The acquisition platform's separate terms will still require verification if applicable.
- LOCKED: OpenStreetMap data will be attributed to OpenStreetMap contributors and its Open Database License will be acknowledged. Publicly distributed maps and databases will follow the applicable OSM attribution and licensing requirements [67].
- [DECISION PENDING: D35 selected-source attribution and redistribution]: If ESA WorldCover will be selected, the team will cite the chosen dataset and use its required attribution; CC BY 4.0 availability was confirmed [61]. The team will define attribution placement and what raw or derived artifacts will be shared. Obligations for any selected elevation data, additional layers, and software will remain [NEEDS VERIFICATION] until those sources will be selected and checked.
- LOCKED: Maps will be described as annual satellite-derived surface thermal estimates for road-associated locations. They will not be presented as a heat-health warning system or as direct evidence of human thermal exposure.
- LOCKED: Reporting will include null or adverse comparative results, missing-data coverage, reference-mask limitations, spatial support, and uncertainty. The interpretation will not claim that road connectivity necessarily improves performance.
- [DECISION PENDING: D36 AI-assistance disclosure]: The team will confirm the department's AI-use policy and determine any required disclosure of assistance in outlining, writing, coding, or editing. Authors will remain responsible for source verification and scientific decisions.
- [DECISION PENDING: D37 ethics-review status]: The team will ask the adviser whether USeP will require an ethics-review exemption statement or another institutional determination for this public-data study. No exemption will be assumed. The applicable institutional policy will remain [NEEDS VERIFICATION].

**Figures and Tables**

- LOCKED: Figure 3.1 will present the conceptual relationships among historical node attributes, alternative adjacency representations, LST estimates, research-question comparisons, and downstream SUHI intensity. Its final visual form will remain pending under D1.
- LOCKED: Figure 3.2 will present the methodological workflow, with separate predictor, target, and non-urban reference branches and a gate before final evaluation. Its detail will remain pending under D3.
- LOCKED: Figure 3.3 will show graph construction from OSM roads to intersection and intermediate nodes, with road adjacency, self-loops only, and spatial-proximity graph panels. Illustrative geometry will not imply selected spacing or k.
- LOCKED: Figure 3.4 will show the spatial split, shared satellite support, buffers or blocks, and the boundary message-passing policy. Exact geometry will remain pending under D19 and D25–D27.
- LOCKED: Table 3.1 will compare the four configurations using the schema below. Table 3.2 will document datasets, years, resolution or support, scaling, QA, provenance, and attribution. Table 3.3 will document partition counts and coverage. Table 3.4 will provide a planned results schema for LST metrics and paired differences without fabricated values.

| Configuration | Historical predictors | Adjacency | Question |
|---|---|---|---|
| 1 LSTM-GraphSAGE | LST, NDVI, NDBI, 2019–2024 | OSM road-connected | RQ1; RQ2; RQ3 |
| 2 Matched LSTM-GraphSAGE | Same as configuration 1 | Self-loops only | RQ2 |
| 3 Matched LSTM-GraphSAGE | Same as configuration 1 | Spatial-proximity graph; rule pending | RQ2 |
| 4 Random Forest | Same historical information; tabular encoding pending | No graph aggregation | RQ3 |

**Source List**

Existing [47] will be retained. New [57]–[68] will be provisional until bibliography reconciliation. Undated living documentation will use n.d. and an access date rather than an invented publication year. VERIFIED-VIA-SEARCH will mean that the linked official source was retrieved; it will not certify an unselected implementation.

[47] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive Representation Learning on Large Graphs,” NIPS, 2017; arXiv revision, 2018. https://arxiv.org/abs/1706.02216. Will support

GraphSAGE aggregation and the distinction between inductive capability and experimental protocol. VERIFIED-VIA-SEARCH.

[57] U.S. Geological Survey, “Landsat 8–9 Collection 2 Level 2 Science Product Guide,” LSDS-1619, version 6, 2024. https://www.usgs.gov/media/files/landsat-8-9-collection-2-level-2-science-product-guide. Will support product structure, band interpretation, validity, and processing documentation. VERIFIED-VIA-SEARCH.

[58] U.S. Geological Survey, “How do I use a scale factor with Landsat Level-2 science products?,” n.d., accessed 11 October 2026. https://www.usgs.gov/faqs/how-do-i-use-a-scale-factor-landsat-level-2-science-products. Will support temperature and reflectance scale factors and offsets. VERIFIED-VIA-SEARCH.

[59] U.S. Geological Survey, “Landsat Collection 2 Quality Assessment Bands,” n.d., accessed 11 October 2026. https://www.usgs.gov/landsat-missions/landsat-collection-2-quality-assessment-bands. Will support QA_PIXEL and QA_RADSAT interpretation without selecting screening thresholds. VERIFIED-VIA-SEARCH.

[60] ESA WorldCover consortium, “WorldCover Product User Manual,” version 2.0, 2022. https://esa-worldcover.s3.eu-central-1.amazonaws.com/v200/2021/docs/WorldCover_PUM_V2.0.pdf. Will support interpretation of the 2021 v200 land-cover product if selected. VERIFIED-VIA-SEARCH.

[61] ESA WorldCover project, “Data,” n.d., accessed 11 October 2026. https://esa-worldcover.org/en/data-access. Will support available product editions, dataset citations, CC BY 4.0 licensing, and attribution. VERIFIED-VIA-SEARCH.

[62] OSMnx project, “User Reference” and “Getting Started,” living documentation, n.d., accessed 11 October 2026. https://osmnx.readthedocs.io/en/stable/user-reference.html and https://osmnx.readthedocs.io/en/stable/getting-started.html. Will support graph extraction, road filters, simplification, and component retention. The observed documentation release will not be an installed-version claim. VERIFIED-VIA-SEARCH.

[63] PyTorch Geometric project, “torch_geometric.nn.conv.SAGEConv,” living documentation, n.d., accessed 11 October 2026. https://pytorch-geometric.readthedocs.io/en/latest/generated/torch_geometric.nn.conv.SAGEConv.html. Will support aggregation and root-feature implementation decisions. VERIFIED-VIA-SEARCH.

[64] scikit-learn developers, “RandomForestRegressor,” living documentation, n.d., accessed 11 October 2026. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html. Will support baseline implementation and hyperparameter documentation. VERIFIED-VIA-SEARCH.

[65] scikit-learn developers, “Metrics and scoring: quantifying the quality of predictions,” living documentation, n.d., accessed 11 October 2026. https://scikit-learn.org/stable/modules/model_evaluation.html. Will support regression metric definitions and reporting conventions. VERIFIED-VIA-SEARCH.

[66] U.S. Geological Survey, “Landsat 8” and “Landsat Collection 2 Surface Temperature,” n.d., accessed 11 October 2026. https://www.usgs.gov/landsat-missions/landsat-8 and https://www.usgs.gov/landsat-missions/landsat-collection-2-surface-temperature. Will support the nominal 16-day repeat cycle, native thermal sampling, and temperature-product limitations. VERIFIED-VIA-SEARCH.

[67] OpenStreetMap contributors and OpenStreetMap Foundation, “Copyright and License,” n.d., accessed 11 October 2026. https://www.openstreetmap.org/copyright. Will support ODbL and attribution requirements. VERIFIED-VIA-SEARCH.

[68] U.S. Geological Survey, “Are Landsat data in the cloud still considered to be within the public domain?,” n.d., accessed 11 October 2026. https://www.usgs.gov/faqs/are-landsat-data-cloud-still-considered-be-within-public-domain. Will support Landsat public-domain status. VERIFIED-VIA-SEARCH.

**Pending Decisions Table**

Each row will identify a team choice rather than a recommendation. Options will remain subject to the locked design constraints; the team will record its selection before dependent procedures will be implemented.

| Item | Why it matters | Options and trade-offs | RQ |
|---|---|---|---|
| D1 Framework style | Will meet department expectations. | Input-process-output: clear flow; relationships diagram: clearer comparisons; theory-based: stronger literature linkage. | All |

| | | | |
|---|---|---|---|
| D2 Literature and citations | Will align Chapters I–III. | Retain verified entries: stable numbering; reconcile and renumber: cleaner list. Both will correct duplicate subsection labels and missing references. | All |
| D3 Workflow figure | Will expose processing dependencies. | Single overview: compact; overview plus leakage panel: more explicit controls. | All |
| D4 Training and graph population | Will define context and generalization. | Urban-only graph: focused scope; expanded context graph: wider information, more boundary controls. | All |
| D5 Urban mask | Will define evaluated locations. | Built-up class membership: simple; built-up fraction threshold: support-aware; objective contiguous-area rule: closer to continuous urban extent. Dataset and year will be selected. | All |
| D6 Reference mask | Will shape SUHI interpretation. | Fixed terrestrial ring: reproducible geography; matched non-urban polygons: better comparability; broader eligible terrestrial mask: more coverage. Classes and thresholds will be fixed. | Mapping |
| D7 Elevation comparability | Will avoid a mountain-driven contrast. | Elevation-range restriction: excludes mismatches; matched elevation distribution: stronger comparability; stratified reporting: exposes remaining differences. DEM will be selected. | Mapping |
| D8 Boundary and grid | Will control spatial alignment. | Local projected CRS: meaningful distances; approved regional projected CRS: wider interoperability. Boundary | All |

| | | | |
|---|---|---|---|
| | | snapshot and raster origin will be recorded. | |
| D9 Data acquisition | Will establish provenance. | USGS downloads: explicit files; cloud processing: scalable operations; archived prepared inputs: easier reruns with provenance requirements. | All |
| D10 QA rules | Will balance coverage and reliability. | Conservative QA: fewer valid pixels; product-validity plus documented confidence rules: more coverage; prespecified QA sensitivity analysis: greater workload. No numeric thresholds will be assumed. | All |
| D11 Scaling and units | Will preserve physical meaning. | Kelvin reporting: native physical unit; Celsius reporting: familiar interpretation. Both will use documented product conversion and guard against double scaling. | All |
| D12 Index aggregation | Will define annual predictors. | Median of scene-level indices: summarizes valid ratios; index from median reflectance: summarizes a composite spectrum. Exact bands and denominator rules will be specified. | All |
| D13 Missing values | Will affect coverage and bias. | Complete-case exclusion: simpler, less coverage; training-fitted imputation: retains nodes, adds assumptions; missingness-aware encoding: more complex matched implementation. | All |

| | | | |
|---|---|---|---|
| D14 OSM date and filters | Will define represented roads. | Historical snapshot: closer temporal alignment; current snapshot: accessible, temporal mismatch. Broad versus restricted road classes will trade coverage for scope. | All |
| D15 Node construction | Will affect density and adjacency. | Retain raw intermediate vertices: irregular spacing; densify simplified edges: controlled spacing; hybrid retention: more complex provenance. Spacing will remain unset. | All |
| D16 Adjacency conventions | Will define neighbor exchange. | Undirected road adjacency: symmetric relation; directed adjacency: preserves selected direction rules. Parallel-edge collapse and self-loop treatment will be consistent. | RQ1, RQ2 |
| D17 Components | Will affect representativeness. | Retain all eligible components: broader coverage; main component only: simpler, narrower population; prespecified size rule: compromise with exclusion reporting. | All |
| D18 Raster assignment | Will define the observation at a node. | Containing cell: traceable support; interpolation: smoother values, overlapping support; neighborhood summary: broader context and dependence. | All |
| D19 Support groups | Will prevent shared-observation leakage. | Group identical source cells: addresses duplicates; group contributor footprints with buffers: stronger control; conservative support zones: simpler, fewer samples. | All |

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| D20 Metric weighting | Will define whose errors dominate. | Equal nodes: road-node estimand; equal source pixels: reduces densification dominance; both: clearer sensitivity with more reporting. | All |
| D21 Proximity graph | Will define the RQ2 comparator. | KNN: controlled neighbor count, variable reach; radius graph: controlled reach, variable degree. k or radius, ties, symmetry, and split scope will be selected. | RQ2 |
| D22 Neural architecture | Will control model capacity. | Compact shared architecture: feasible computation; larger shared architecture: more capacity and overfitting risk; validation-selected shared architecture: more search cost. | All |
| D23 Self-information | Will preserve the matched ablation. | Explicit self-loops plus root term: two self contributions; explicit self-loops with root term disabled: one aggregation path. Each policy will apply consistently to configurations 1–3. | RQ1, RQ2 |
| D24 Random Forest settings | Will define a fair baseline. | Year-specific raw columns with fixed settings: transparent; same raw columns with validation tuning: stronger baseline at greater cost. Tree settings will remain unset. | RQ3 |
| D25 Spatial split | Will define spatial generalization. | Spatial blocks: broad sampling; buffered held-out areas: clearer separation; separated geographic groups: interpretable regions. Sizes, proportions, and repeats will remain unset. | All |

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| D26 Training protocol | Will define permitted held-out access. | Transductive: specified historical features and topology visible; inductive: held-out nodes or areas absent during fitting. Neither will permit held-out labels in training. | All |
| D27 Boundary edges | Will control message passing. | Cut crossing edges: stronger separation; buffer constrained neighborhoods: spatial protection; declared transductive exchange: realistic available-feature access with narrower claims. | RQ1, RQ2 |
| D28 Preprocessing and refit | Will protect final evaluation. | Train-only fitting and checkpoint: simple separation; train-plus-validation refit after freezing: more fitting data, additional provenance. Test data will remain untouched. | All |
| D29 Training parameters | Will make training reproducible. | Full-graph training: simple, memory intensive; sampled mini-batches: scalable, more sampling controls. Loss, optimizer, rate, epochs, regularization, and seeds will be selected. | RQ1, RQ2 |
| D30 Tuning procedure | Will keep adjacency comparisons interpretable. | Fixed common neural settings: strict matching; common validation search yielding one shared setting set: more work. Baseline search budget will be prespecified. | All |
| D31 Statistical comparison | Will account for dependence. | Paired spatial-block resampling: spatial uncertainty; paired repeated spatial splits: variable geographic difficulty; descriptive differences: fewer inferential assumptions. | RQ2, RQ3 |

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| D32 Aggregation and diagnostics | Will govern metric interpretation. | Pooled predictions: single summary; partition-level metrics: geographic variation; both: fuller reporting. Seed summaries and metric priority will be defined. | All |
| D33 Maps | Will avoid overstating validation. | Held-out-only maps: visibly validated scope; subsequent full-coverage maps: broader interpretation with distinct labeling. Continuous versus classified symbols will remain pending. | Mapping |
| D34 Software and archive | Will support reproduction. | Local scripts and locked environment: direct control; cloud preprocessing plus local modeling: scalable data preparation; containerized workflow: portable with setup cost. | All |
| D35 Attribution and release | Will meet selected-source obligations. | Code and manifests: smaller release; code and permitted derived assets: easier reproduction; broader licensed archive: more legal and storage checks. | All |
| D36 AI disclosure | Will meet academic policy. | Required acknowledgment: concise transparency; methods appendix or contribution record: more detail. Adviser guidance will determine the compliant form. | All |
| D37 Ethics status | Will meet USeP requirements. | Exemption request if required: documented determination; other institutional review if required: fuller review. Adviser confirmation will precede any exemption claim. | All |

**Risks**

- [NEEDS VERIFICATION]: Chapter I's statement that excluding Landsat 9 will preserve a "mathematically consistent" 16-day sampling frequency and prevent artificial warming biases will require revision or supporting evidence. A nominal orbital repeat cycle will not establish uniform valid observation intervals after cloud screening, path overlap, and missing data [66]. The Landsat 8-only choice will remain locked.
- [NEEDS VERIFICATION]: Chapter I's early references to "street-level" and high-resolution heat mapping will require qualification against its later spatial-support limitations. Road-node placement will not establish street-scale independent thermal measurement. Chapter III will retain the narrower road-associated interpretation [57], [66].
- [NEEDS VERIFICATION]: Citation [56] will require a bibliography entry or corrected citation. Chapter I's [38] for spatial-dependence evaluation appears to point to a thermal-landscape graph study, so its suitability will require checking. The paired citations [41][45] and [42][46] will require checking against their actual subjects. Chapter II's repeated 2.1.1 numbering and bracketed title placeholders will require correction.
- [DECISION PENDING: D2 literature linkage]: Chapter II's language about isolating thermal impacts and preventing peri-urban contamination will be reviewed for compatibility with the non-causal comparative estimation design. Reference geometry alone will not establish causal isolation. The framework will avoid importing physical heat-transfer claims from unrelated graph applications.
- [DECISION PENDING: D4–D8 and D14–D18 operational definitions]: Undefined "urban," "continuously built-up," "non-urban reference," "intermediate point," "road-connected," and "spatial-proximity" will be resolved through documented masks, graph rules, and observation-support definitions rather than visual judgment or desirable model errors.
- LOCKED: Shared target pixels will create leakage if their nodes will be split across training and test partitions. Resampling or interpolation will also permit overlap beyond identical delivered pixel IDs. The grouping and buffer design will address these cases before training.
- LOCKED: Cross-boundary graph edges will permit information exchange beyond split geometry. The protocol will distinguish declared access to historical held-out features from prohibited access to held-out 2025 labels, and will account for multi-layer neighborhoods.
- LOCKED: Scaling, imputation, feature selection, normalization, and tuning will leak information if fitted using validation or test data contrary to the selected protocol. Test-based mask adjustment, k selection, graph filtering, or residual-based node exclusion will invalidate the held-out comparison.
- LOCKED: The 2025 training labels will make the study a retrospective spatial estimation experiment. Good performance will not establish temporal generalization to later years. The 2025 reference will be used only after prediction for mapping and will not support an operational forecast claim.
- LOCKED: A shared reference subtraction will not provide an independent second prediction-error experiment. SUHI intensity will therefore remain a derived mapping output, and model comparison will remain on LST.
- [DECISION PENDING: D19–D20 and D31–D32 comparison validity]: Dense road sampling will inflate nominal sample counts and may overweight certain areas. Treating nodes or repeated seeds as independent spatial replicates will overstate statistical precision. Inference will follow the selected support grouping and spatial comparison procedure.
- LOCKED: Differences between road-connected and spatial-proximity graph performance will concern their complete adjacency definitions, including neighbor counts and spatial reach. They

will not isolate a causal effect of individual road links. Null results and poorer performance will remain reportable outcomes.